

Fig. 8: An actual-size, single-side PCB layout for vehicle power adaptor

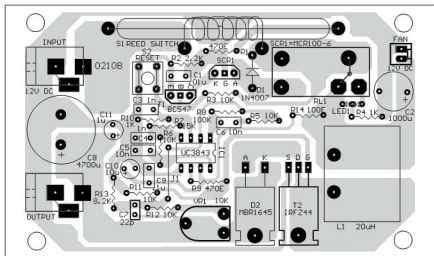


Fig. 9: Component layout for the PCB

eration of IC1. The output frequency from pin 6 of IC1 drives the gate of power MOSFET T2 for switching action to the input power supply. Fig. 7 shows the pin configurations of BC547, MBR1645 and IRFZ44.

Resistor R11 turns MOSFET T2 'off' if for any reason resistor R9 becomes

open or IC1 is defective. Electrolytic capacitor C8 is used to smooth the output and capacitor C11 is used for high-frequency decoupling.

The inductor coil (L1) is made upon EE core from several lengths of an enamelled wire connected in parallel to reduce the skin effect, which becomes a problem with high switching speed. Schottky diode D2 has a TO-220 package.

In normal operation, the energy is fed from inductor L1 to the load, then stored in the output capacitor. For this reason, the output capacitor is stressed.

A simple reed switch (S1) is used here to detect the over-current or short-circuit condition. About ten

turns of 18 SWG over the reed switch are sufficient to detect the over-current condition.

Initially, when the circuit is powered on, SCR1 is off. The DC-to-DC converter is connected through normally-open (N/C) contacts of relay RL1. When the current drawn by the load increases to a certain limit, the contacts of the reed switch close to trigger SCR1. The relay energises and the load gets disconnected. Once the short circuit is removed, the circuit can be reset by pushing switch S2. The bicolour LED indicates converter working as well as faulty mode.

Construction and testing

An actual-size, single-side PCB for the vehicle power adaptor is shown in Fig. 8 and its component layout in Fig. 9. Because of the high-current switch-mode power supply, it is advised to assemble the circuit on a PCB. The PCB minimises time and assembly errors. Carefully assemble the components and double-check for any overlooked error. Mount power MOSFET IRFZ44 and Schottky diode D2 using suitable heat-sinks.

For testing, use either a 12V, 10A power supply or 12V vehicle battery. When no obvious malfunction occurs with the power connected, slowly adjust potentiometer VR1 to ensure that it can achieve an output voltage of 19V. Now use a high-power resistor (8 ohms, 10W) and adjust potentiometer VR1 for 19V output for testing.

The finished unit can be housed in a cabinet. A long usage requires ventilation for heat dissipation. Sufficient air-circulation is required to safeguard the power-handling electronics in the circuit. A few holes in the case may help but it is better to have a 3x3x1cm³ cooling fan.

Caution

Make sure that the cigarette lighter socket and the plug are capable of handling 10A current. Connect the adaptor to the laptop with a suitable lead and connector. Double-check the polarity of the input pins before plugging to avoid damage to the laptop. ●

Inductor Specifications

Parameter	Values
Inductance value	20 μ H
Core material	Epoxy N27 or equivalent
Core size	E core 25/13/7
Core gap	To be adjusted to get 20 μ H
Number of turns	15
Enamelled copper wire	10 strands of 30 SWG